

Disaster Recovery and Business Continuity Plan

Site Reliability Engineering | Version 4.2 | Last reviewed Q1 FY2026-27

INTERNAL — Engineering and SRE Teams

1. Scope, Authority, and Activation Triggers

This Disaster Recovery and Business Continuity Plan covers the Nova Platform production environment, the customer-facing API surface, the AI/ML inference layer, and all primary data stores. It does not cover individual employee workstations or non-production environments.

Authority to declare a disaster rests jointly with the Chief Technology Officer (Vikram Reddy) and the Chief Information Security Officer (Karthik Iyer); a single declaration by either is sufficient. In their absence, the SVP of Engineering or the Head of Site Reliability Engineering may declare. All declarations must be communicated to the Board IPO Committee within 24 hours.

Activation triggers include: (i) a SEV-1 incident exceeding two hours of unmitigated impact; (ii) primary AWS region (ap-south-1) unavailability exceeding 30 minutes; (iii) confirmed compromise of customer-managed encryption keys (BYOK); (iv) regulatory order to suspend service in any jurisdiction; or (v) physical damage at the Bangalore office that prevents Engineering and SRE teams from operating.

2. Recovery Time and Recovery Point Objectives

The platform is segmented into three service tiers with distinct recovery targets, aligned to the `criticality_tier` values in the `products_services` table.

Tier 1 (Critical) services: RTO of 30 minutes, RPO of 5 minutes. This applies to user-management, authentication, the API gateway, and the core analytics engine. These are the services whose total outage constitutes a SEV-1 event.

Tier 2 (High) services: RTO of 2 hours, RPO of 30 minutes. Includes the data-processing batch pipelines, the AI/ML inference layer, and the integration hub.

Tier 3 (Medium and Low) services: RTO of 8 hours, RPO of 4 hours. Includes administrative tooling, internal dashboards, and the reporting subsystem.

These RTO/RPO commitments are stricter than the contractual SLA in customer MSAs by design; the customer SLA of 99.97% uptime translates to a permissible outage budget of 2.6 hours per quarter, which the internal targets aim to keep well below.

The contractual SLA cap on aggregate service credits in any twelve-month period is 20% of the affected customer's annual fees. Service credit floor varies by contract: standard is 5%, premium-tier customers negotiate up to 20%.

3. Disaster Recovery Architecture

The platform runs on AWS as primary (ap-south-1 region in Mumbai) with Google Cloud Platform (asia-southeast1 in Singapore) for disaster recovery. Replication is asynchronous for transactional databases (PostgreSQL 16 on AWS RDS Multi-AZ) with a target replication lag of less than 30 seconds during normal operation.

Read replicas are deployed in three additional AWS regions: us-east-1, eu-west-1, and ap-south-1. Time-series telemetry and audit logs are stored in TimescaleDB with a 90-day hot retention window and S3-backed cold storage for 7-year compliance retention. Search indices for customer-facing analytics are maintained on Elasticsearch (18 nodes, 142 TB), with cross-region replication to the GCP DR site.

The AI/ML production stack runs on a dedicated single EKS cluster with 16 NVIDIA A100 GPU nodes (8 GPUs per node = 128 GPUs total). Critical limitation: there is no regional redundancy for AI workloads, meaning AI inference cannot failover to GCP. The FY26-27 product roadmap allocates capital for a Tokyo and Frankfurt AI failover architecture; until that completes, AI is a single point of failure.

4. Failover Procedures by Service Tier

For Tier 1 service failover, the runbook prescribes the following sequence. Step 1: SRE primary on-call validates the trigger (RTO clock starts at trigger confirmation). Step 2: Initiate AWS to GCP failover via the pre-staged Terraform pipeline; this is a 18-minute median operation per quarterly drill records. Step 3: Update the global DNS (Route 53 weighted records) to redirect customer traffic to the GCP endpoint. Step 4: Validate authentication, API gateway, and analytics-engine health from synthetic monitoring within 5 minutes of DNS propagation. Step 5: Notify customers via the status-page and email. Step 6: Post-incident review within 5 business days.

For Tier 2 service failover, additional steps include rebuilding the data-processing batch pipelines from the last completed checkpoint (RPO of 30 minutes means up to 30 minutes of pending work may need to be re-run). For the AI/ML inference layer specifically, since regional failover is not yet available, the documented procedure is to fail OPEN in degraded mode — serving cached predictions from a 24-hour Redis tier — for up to 4 hours while the EKS cluster is restored.

For Tier 3 services, manual failover is acceptable and SRE on-call may defer to business-hours.

5. DR Drill History and Findings

DR drills are conducted quarterly. The last four drills are summarised below:

Q1 FY2025-26 drill (June 2025): Tier 1 failover. Achieved RTO of 28 minutes (target: 30 minutes). PASS. Two findings: (i) DNS propagation delay observed in ap-southeast-2 region (5 minutes); (ii) authentication token validation latency increased by 40% post-failover until cache warmed. Both remediated by Q2.

Q2 FY2025-26 drill (September 2025): Full-stack DR including Tier 2. RTO 2 hours 18 minutes (target: 2 hours). FAIL. Findings: (i) data-processing pipeline restart required manual intervention not in the runbook; (ii) one customer-managed encryption key rotation was not handled cleanly. Both remediated by Q3.

Q3 FY2025-26 drill (December 2025): Tabletop-only following INC-2025-0847; full failover deferred to Q4. Findings focused on incident-response coordination with security team.

Q4 FY2025-26 drill (March 2026): Tier 1 failover. RTO 22 minutes. PASS with no findings. The next full drill is scheduled for June 2026.

6. Crisis Communication and Escalation

During an active disaster, communication is governed by a strict matrix to avoid contradictory messaging.

Internal communication: a Slack incident channel is auto-created by the on-call paging tool; the SRE on-call commander chairs it. The incident channel must include representatives from Engineering, SRE, Customer Success, Legal, and Communications. The Board IPO Committee is notified within 24 hours of any declaration.

External customer communication: the status page is the canonical source; updates are posted within 15 minutes of declaration and every 30 minutes thereafter until resolution. For Tier-1 customers and Tier-2 customers in regulated industries (BFSI, Healthcare), Customer Success initiates direct outreach within 1 hour. The CEO and CRO must approve any customer communication that names a specific cause until the post-incident review confirms the root cause.

Regulatory communication: for any breach involving customer personal data, CERT-In must be notified within 6 hours. For DPDP Act personal data breaches, the Data Protection Board must be notified. Legal counsel (Khaitan & Co) is engaged automatically for any regulatory communication.

Media communication: only the CEO and the Head of Communications are authorised to engage media. All media enquiries during an active disaster are directed to communications@technova.com with a default no-comment policy until cleared.